



# GARASU

BETA

GLASS AGING, RELAXATION AND SIMULATION UTILITY

## User Guide



Beta

Learn

Lab

Fitter

Data Analysis

### GARASU

Glass Aging, Relaxation And Simulation Utility Beta

Project **none** ▾ New project

Every glass carries the **thermal fingerprint** of how it was made, written in its fictive temperature. GARASU simulates and fits that history, using the heterogeneous **Takeda–Lucas** model alongside **TNM** and **RelaxPy**, for DSC, MDSC, enthalpy, volume and annealing data.

Start learning Analyze data Open Fitter Download for Mac / Windows

#### Tools

##### Learn

Experiments & models

How structural relaxation is measured, and what the TNM, MAP and Takeda–Lucas models assume, with live simulations throughout.

- Annealing, Kovacs asymmetry and memory, DSC, enthalpy recovery, MDSC  $C_p^*$ , reversing/non-reversing and their caveats
- The mathematics of TNM, MAP (RelaxPy) and TL
- Mean field vs Prony vs domain  $T_f$ : see the distributions

[Open Learn](#)

##### Data Analysis

Your data

Turn raw DSC heat flow and relaxation data into normalized curves, fictive temperatures and enthalpy.

- Glass and liquid baselines  $\rightarrow C_p^N, T_f(T), T_f'$  for heating and cooling scans
- Heating after annealing: aligned scans,  $\Delta H_f$ , recovered  $\Delta H(t_0)$ ,  $\phi$  and a KWW fit
- Volume / density / length during annealing: stretched-exponential fit

[Open Data Analysis](#)

##### Fitter

Step by step

Program any thermal history, load your measurements, then compute with chosen parameters or fit them.

- Ramps, anneals with log-spaced times, T-jumps, MDSC modulation
- DSC, MDSC  $C_p^*/C_p^*$ , enthalpy, volume,  $T_f$ , annealing data
- Bounded local / global fits, errors, correlations, AIC/BIC

[Open Fitter](#)

Documentation: [Quick Start \(PDF, 2 pages\)](#) · [User Guide \(PDF, 25 pages\)](#)

Hands-on: the **Lab** runs cool–heat scans, anneals and T-jumps with all three models side by side, with relaxation-time distributions and  $\beta_{KWW}(T)$ .

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## In a hurry?

The separate **Quick Start** document takes you from installing to a first result in a few minutes. This guide explains every screen and option in detail.

# 1 • What GARASU does

Every glass carries a thermal fingerprint of how it was made, written in its fictive temperature  $T_f$ . GARASU simulates that history, analyzes your measurements, and fits relaxation models to them.

GARASU contains four tools. The red bar on the left of every page switches between them, on the web and in the desktop app; the icon at its top returns to the home page.

| Tool          | Use it to...                                                                                                                                                                |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learn         | understand the experiments (annealing, Kovacs jumps, DSC, enthalpy recovery, MDSC) and the models (TNM, MAP / RelaxPy, Takeda–Lucas), with live simulations on every topic. |
| Lab           | try a cool–heat scan, an anneal or a T-jump interactively and compare the three models side by side.                                                                        |
| Fitter        | program any thermal history, load your data, then compute with chosen parameters or fit the parameters, with uncertainties.                                                 |
| Data Analysis | turn raw DSC heat flow into normalized $C_p$ , fictive temperatures and recovered enthalpy, and fit volume (or density, length...) relaxation during annealing.             |

## The three models

| Model                             | In one line                                                                                                                                                                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Takeda–Lucas (TL)                 | Heterogeneous domains, each with its own Vogel temperature and local fictive temperature $T_{fi}$ ; the spread of $T_{fi}$ persists in equilibrium, giving temperature-dependent nonexponentiality and enthalpy fluctuation (J. Chem. Phys. 160, 174504, 2024). |
| Tool–Narayanaswamy–Moynihan (TNM) | One fictive temperature for the whole sample, nonlinearity $x$ and a stretched-exponential (KWW) kernel.                                                                                                                                                        |
| MAP / RelaxPy                     | Mauro–Allan–Potuzak nonequilibrium viscosity with Prony-series components (Wilkinson, Mauro & Mauro, SoftwareX 2018).                                                                                                                                           |

### Your data stay on your computer

All calculations run locally, in your browser or in the desktop app. Nothing you load is uploaded anywhere.

### Beta software

GARASU is in beta: features, defaults and occasionally results may change between versions. Record the version number (bottom of the home page in the desktop app) with your results.

## 2 · Getting GARASU

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### 2.1 In the browser

Open [wtakeda-research.github.io/GARASU](https://wtakeda-research.github.io/GARASU). Nothing to install; it works in Chrome, Edge, Firefox and Safari, and always shows the latest version. After the page has loaded, the tools also work offline.

### 2.2 Desktop app

Download the installer from [github.com/wtakeda-research/GARASU/releases/latest](https://github.com/wtakeda-research/GARASU/releases/latest), or click **Download for Mac / Windows** on the home page.

#### macOS (Apple silicon and Intel)

- 1 Download `GARASU-<version>-universal.dmg`.
- 2 Open it and drag **GARASU** into **Applications** (choose **Replace** if asked).
- 3 Open GARASU. The first time, macOS says it cannot verify the developer: click **Done**.
- 4 Open **System Settings** → **Privacy & Security**, scroll down and click **Open Anyway**, then confirm.

This is needed once per computer, because the app is signed with the group's own certificate rather than an Apple Developer ID.

#### Windows 10 / 11

- 1 Download `GARASU-Setup-<version>.exe`.
- 2 Run it. If Windows shows “Windows protected your PC”, click **More info** → **Run anyway**.
- 3 Choose the install folder (no administrator rights are needed) and finish.
- 4 Start GARASU from the Start menu or the desktop shortcut.

### 2.3 Updates

The desktop app checks for a new version when it starts and every six hours; you can also check with **GARASU** → **Check for Updates...** (Mac) or **Help** → **Check for Updates...** (Windows). A new version downloads in the background and the app offers **Restart now** to install it.

If macOS cannot install an update in place, the app says so and offers **Download**, which opens the release page: install the new DMG as in 2.2. The web version needs no updates.

### 2.4 One window for every tool

The desktop app opens a single window on the home page. The bar on the left switches tools inside that window, and each tool stays as you left it: a computed Fitter run is still there after a visit to Learn. Shortcuts: `⌘/Ctrl 1` Learn, `2` Lab, `3` Fitter, `4` Data Analysis, `⌘/Ctrl 0` home. The version number is shown at the bottom of the home page.

### 3 · Controls shared by every tool



**Figure 3.1.** Header of the Fitter. Top row: tool name, project, contents and settings; second row: the four steps. Data Analysis has the same layout.

| Control                       | What it does                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tool bar (left)               | Learn, Lab, Fitter and Data Analysis; the icon at the top opens the home page. The current tool is highlighted.                                                                                                                                                                                                                 |
| Project ▼ and Contents        | The project you are working in, shared by the Fitter, Data Analysis and the home page (chapter 8).                                                                                                                                                                                                                              |
| Temperature K / °C            | Unit for every temperature on screen: plots, thermal-history inputs, $T_g$ and its bounds, data limits. Values are stored in kelvin, so switching never changes a calculation.                                                                                                                                                  |
| Time axis auto / linear / log | <i>Auto</i> (default): real (linear) time whenever the temperature changes with time (ramps, MDSC), so scans are shown in real time; log time only for isothermal histories (jump + hold). Relaxation during holds and time-domain hold data stay on log time. <i>Linear</i> or <i>log</i> force one scale for every time axis. |
| ☾                             | Switches between the light (default) and dark colour scheme.                                                                                                                                                                                                                                                                    |
| Export ▼ (on each plot)       | Saves that plot as a PNG or JPEG image, an SVG vector graphic, or its curves as CSV (section 8.6).                                                                                                                                                                                                                              |
| Hover on a plot               | Shows the nearest data point and its values.                                                                                                                                                                                                                                                                                    |

**Typing numbers**  
Inputs accept any number format ( `1e5` , `0.5` , `-40` ). Temperatures are read in the unit selected in the header.

## 4 • Learn

Learn is a self-contained course on structural relaxation. The contents on the left jump to any section; **Part I** covers experiments and **Part II** the models.

**Learn**

TL TNM MAP Experiments Models

**Learn: glass relaxation, from experiment to model**

Part I covers the experiments used to study structural relaxation: annealing, Kovacs jumps, DSC, enthalpy recovery and modulated DSC. Each one comes with a live simulation. Part II explains the mathematics of the three models and where they differ. TNM gives the whole sample one fictive temperature. MAP splits the relaxation into Prony components, and all of them return to the same temperature in equilibrium. The Takeda–Lucas model gives every domain its own fictive temperature, so the fictive temperatures stay spread out even in the equilibrium liquid.

**1 Fictive temperature and structural relaxation**

Above the glass transition, a liquid's structure keeps pace with temperature. On cooling, the structural relaxation time  $\tau$  grows so quickly that the structure eventually cannot keep up and freezes. Tool described the frozen structure by a **fictive temperature**  $T_f$ ; the temperature at which the liquid would have this structure in equilibrium. In equilibrium  $T_f = T$ . In the glass,  $T_f$  records the thermal history, which is why GARASU calls it the *thermal fingerprint*.

Enthalpy and volume follow  $T_f$ . With liquid and glass heat capacities  $C_{p,l}$  and  $C_{p,g}$ , the normalized heat capacity measured by DSC is

$$C_p^N(T) = \frac{C_p - C_{p,g}}{C_{p,l} - C_{p,g}} = \frac{dT_f}{dT}.$$

Relaxation towards equilibrium has three well-known signatures, and every model below has to reproduce all three:

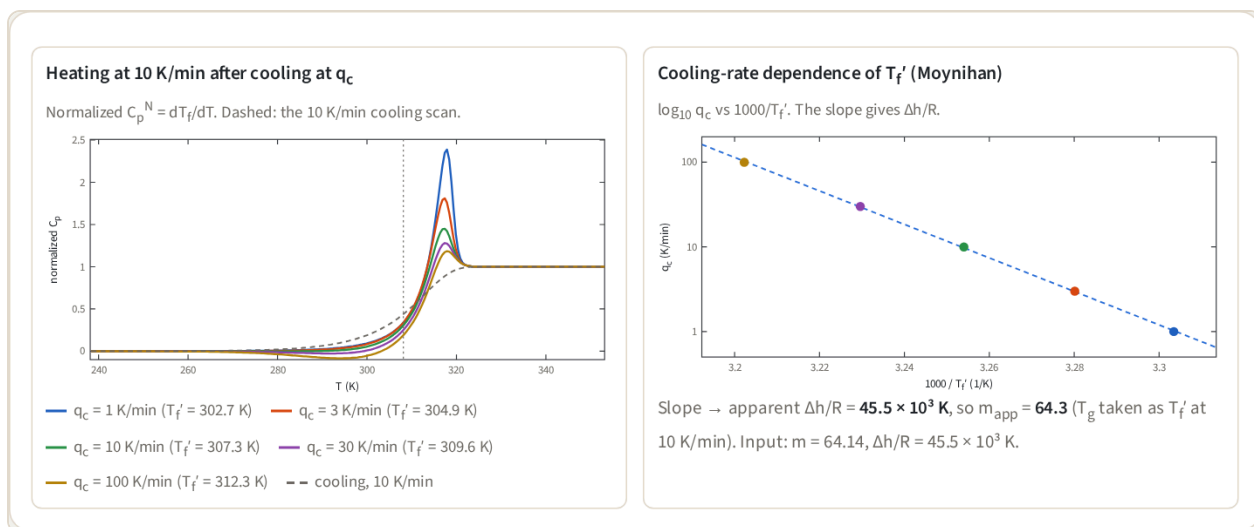
- **Nonexponential.** After a small temperature step, the response follows a stretched exponential (Kohlrausch–Williams–Watts, KWW),  $\Psi(t) = \exp[-(t/\tau)^\beta]$  with  $0 < \beta \leq 1$ .
- **Nonlinear.**  $\tau$  depends on the structure ( $T_f$ ) as well as on  $T$ . A glass therefore relaxes faster when it approaches equilibrium from above than from below (Kovacs asymmetry).
- **Non-Arrhenius.** In equilibrium,  $\tau(T)$  follows a VFT, Adam–Gibbs or MEGA form, and its steepness at  $T_g$  is the fragility  $m = d \log_{10} \tau / d(T_g/T)|_{T_g}$ .

Figure 4.1. Learn. The switch at the top (TL / TNM / MAP) chooses the model used by all simulations in Part I.

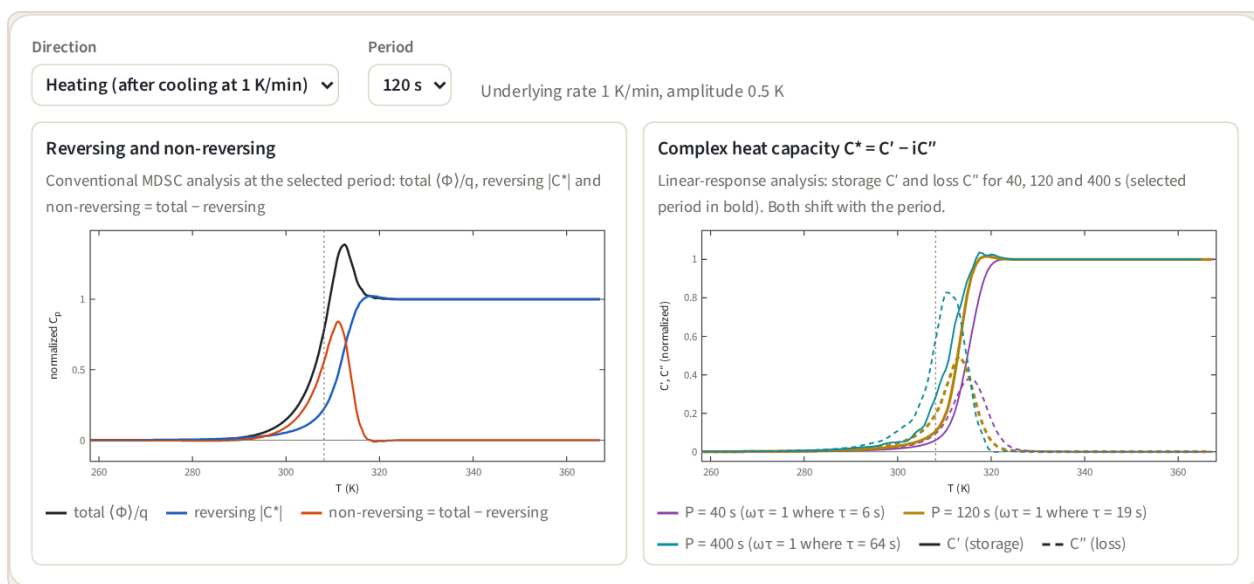
### Part I • Experiments

Each experiment has its protocol and equations, then a live simulation of the same example glass, selenium ( $T_g = 308.13$  K,  $m = 64.14$ , with the TL parameters of Takeda & Lucas 2024; for MAP the glassy viscosity is Arrhenius with  $B = x m T_g$ ,  $x = 0.5$ , and  $C = 0$ ):

- **Isothermal annealing:** relaxation function  $\phi(t)$  after a 10 K down-jump, compared with a linear KWW curve, and the apparent KWW parameters.
- **Kovacs experiments:** asymmetry of approach ( $\pm 5$  K jumps) and the memory effect (Kovacs; Macedo–Napolitano crossover).
- **DSC on cooling and heating:** heating scans after five cooling rates and Moynihan's cooling-rate plot (apparent  $\Delta h/R$  and fragility).
- **Enthalpy recovery:** heating scans after annealing and  $\Delta H(t_a)$ .
- **MDSC:** total, reversing and non-reversing signals,  $C''$ , and how the reversing step moves with the modulation period (the Hutchinson and Schawe caveats).



**Figure 4.2.** DSC section: heating after five cooling rates (left) and the cooling-rate dependence of  $T_f'$  (right).



**Figure 4.3.** MDSC section. Choose **Direction** (heating or cooling) and **Period**. Left: the conventional analysis (total, reversing  $|C^*|$  and non-reversing). Right: the complex heat capacity,  $C'$  (solid) and  $C''$  (dashed), for 40, 120 and 400 s.

## Part II • Models

The mathematics of TNM, MAP and TL, what each assumes, and an interactive comparison of where the fictive temperatures sit.



**Figure 4.4.** “See the difference”: the three models run the same cool–anneal history. Drag the slider or press ► **Play**; change the cooling rate, annealing temperature and TL width  $f$ .

## 5 • Lab

The Lab (formerly Explorer) is a quick, always-live playground: change a number and the plots update. Use it to build intuition; use the Fitter for your own histories and data.

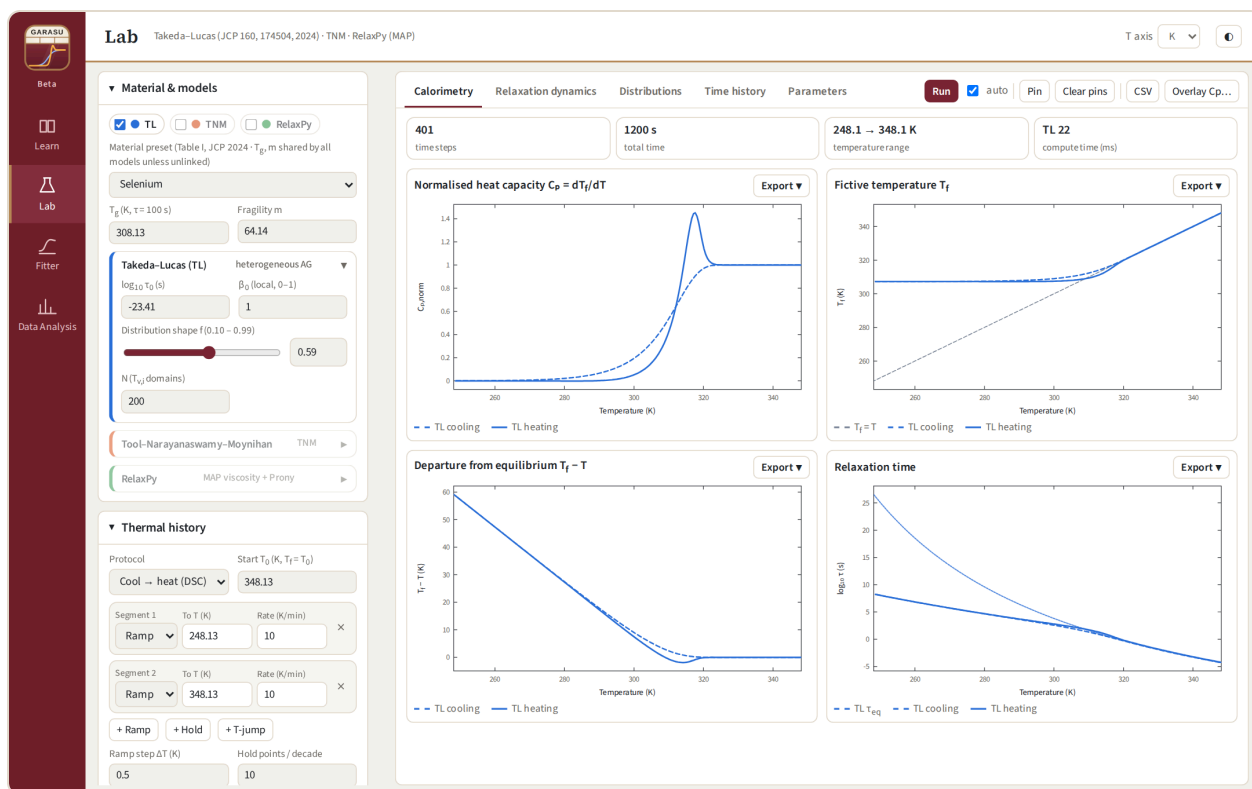


Figure 5.1. Lab: models and material on the left, protocol below; tabs on the right.

- **Models:** tick TL, TNM and/or RelaxPy. **Material preset** loads the TL parameters of Table I of the JCP 2024 paper;  $T_g$  and  $m$  are shared by the models unless unlinked in their sections.
- **Thermal history:** choose a protocol (cool → heat, slow cool, anneal, Kovacs T-jump) or build segments with + Ramp, + Hold, + T-jump.
- **Tabs:** Calorimetry ( $C_p$ ,  $T_f$ ), Relaxation dynamics, Distributions (TL relaxation-time distributions and  $\beta_{KWW}(T)$ ), Time history, Parameters.
- **Run / auto:** untick **auto** to update only when you click **Run**. **Pin** keeps a run for comparison; **CSV** exports the curves.

# 6 • Fitter

The Fitter works in four steps, shown as a switch in the header. You can move between them at any time.

| Step                 | What you do                                                    | Right panel shows                |
|----------------------|----------------------------------------------------------------|----------------------------------|
| 1 Thermal history    | Build the temperature program.                                 | T(t) and the time steps          |
| 2 Experimental data  | Optional: add measurements.                                    | your data and the linked segment |
| 3 Model & parameters | Choose a model, values, which parameters are free, and bounds. | history and data                 |
| 4 Compute & fit      | Click <b>Compute</b> or <b>Run fit</b> .                       | all results                      |

Nothing is calculated until you click

Steps 1–3 only show the thermal history and your data. The model runs when you click **Compute** or **Run fit** in step 4. Any later change clears the old result and asks you to compute again, so a result on screen always matches the current settings.

## 6.1 Step 1 • Thermal history

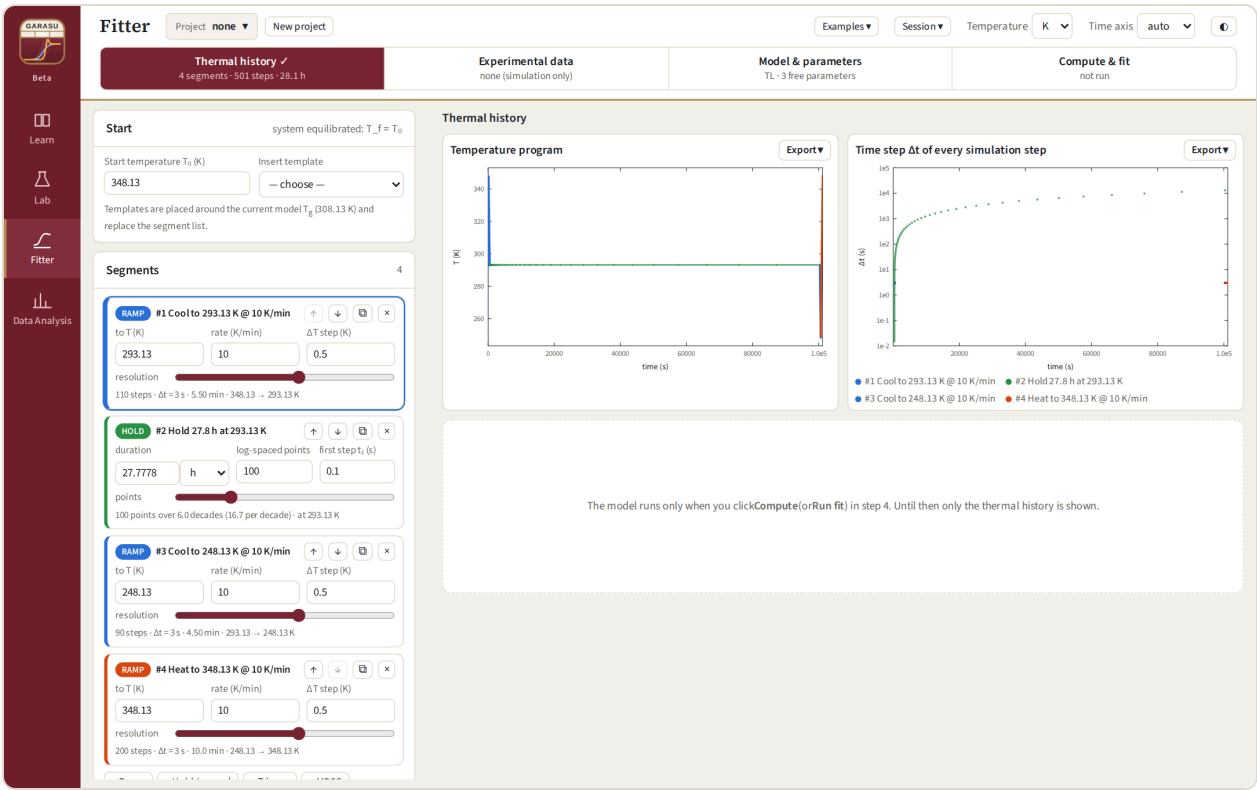


Figure 6.1. Step 1 with the template “Aging below  $T_g$ , then DSC heating”.

- 1 **Start temperature  $T_0$ .** The sample starts in equilibrium at  $T_0$  ( $T_f = T_0$ ). Choose  $T_0$  above  $T_g$ .
- 2 **Insert template** (optional). Templates are placed around the current model  $T_g$ , so set  $T_g$  in step 3 first if your material differs from the default.
- 3 **Add segments with + Ramp, + Hold / anneal, + T-jump, + MDSC.** Reorder, duplicate or delete them with the small buttons on each card.

| Segment       | Inputs                                                                   | Notes                                                                            |
|---------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Ramp          | target T, rate (K/min), $\Delta T$ step                                  | 0.1–0.5 K steps are accurate for DSC; the slider sets $\Delta T$ on a log scale. |
| Hold / anneal | duration (s, min, h, d), log-spaced points, first step $t_1$             | Points are spaced logarithmically from $t_1$ to the duration.                    |
| T-jump        | target T                                                                 | Instantaneous change (e.g. Kovacs experiments).                                  |
| MDSC          | target T, underlying rate, amplitude $\pm A$ , period, points per period | Rate 0 gives quasi-isothermal MDSC with a duration.                              |

Under the segments, the summary shows the number of simulation steps and the total time. The right panel plots the temperature program and the time step of every simulation step.

## 6.2 Step 2 • Experimental data (optional)

The screenshot shows the GARASU Fitter interface. The top bar includes 'Project: none' and 'New project'. The main workspace is divided into three panels. The left panel, 'Datasets', lists two datasets: 'enthalpy relaxation during aging (synthetic)' and 'DSC heating C<sub>p</sub>, raw (synthetic)'. The middle panel, 'Add dataset', is open, showing a form for adding new data. The right panel, 'First rows', displays a table of data points.

| x (s)   | y       |
|---------|---------|
| 3       | 108.402 |
| 4.62993 | 108.401 |
| 7.14542 | 108.397 |
| 11.0276 | 108.349 |
| 17.019  | 108.353 |
| 26.2656 | 108.356 |
| 40.536  | 108.289 |
| 62.5596 | 108.256 |

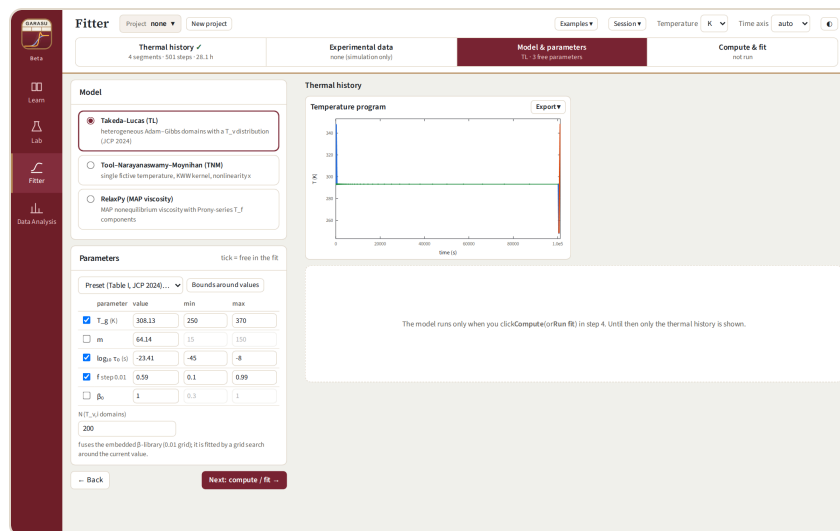
Figure 6.2. Step 2 with two datasets and the Add dataset form open.

- 1 Click + **Add dataset** and choose the **Data type** (table in section 11).
- 2 Choose the **Linked history segment**: the part of the thermal history the data belong to (e.g. the heating ramp for a DSC heating scan, the hold for annealing data).
- 3 **Paste / file**: paste two or more columns or click **Load file...**, then pick the x and y columns and units. Or **Synthesize from model** to generate test data with noise.
- 4 Optional: use  $x \geq / \leq$  to limit the fitted range, a **weight**, a **y multiplier**, and **fit an offset and a scale factor** ( $y = a + b \cdot \text{model}$ ) when the data are not normalized.
- 5 Click **Add dataset**. Tick or untick a dataset to include it;  edits it,  $\times$  removes it.

### Raw data are fine

Types such as “ $C_p$  or heat flow vs T”, “Enthalpy vs T” and “property vs t” fit their baselines or linear coefficients exactly at every step, so you can load raw measurements. To normalize DSC data yourself first, use Data Analysis (section 7).

## 6.3 Step 3 · Model and parameters



**Figure 6.3.** Step 3: model, parameter values, tick = free in the fit, and fit bounds.

- Choose **Takeda–Lucas**, **TNM** or **RelaxPy (MAP)**.
- Type values. Ticked parameters are **free** in a fit and move between **min** and **max**; unticked ones stay fixed. **Bounds around values** sets sensible bounds.
- TL: **Preset** loads Table I of the JCP 2024 paper; **N** is the number of domains (200 is accurate). **f** is fitted on its 0.01 grid.
- TNM: choose the exact KWW summation or the Prony kernel for display (fits always use Prony).
- MAP: **RelaxPy example glass**; A can follow from continuity at  $T_g$ , and  $p = 0.3082 \cdot m$  (RelaxPy default).

## 6.4 Step 4 · Compute and fit

### Compute with chosen parameters

Forward calculation, no fitting. Check or adjust the values and click **Compute**. With data, the agreement ( $R^2$ , RMSE per dataset) is shown.

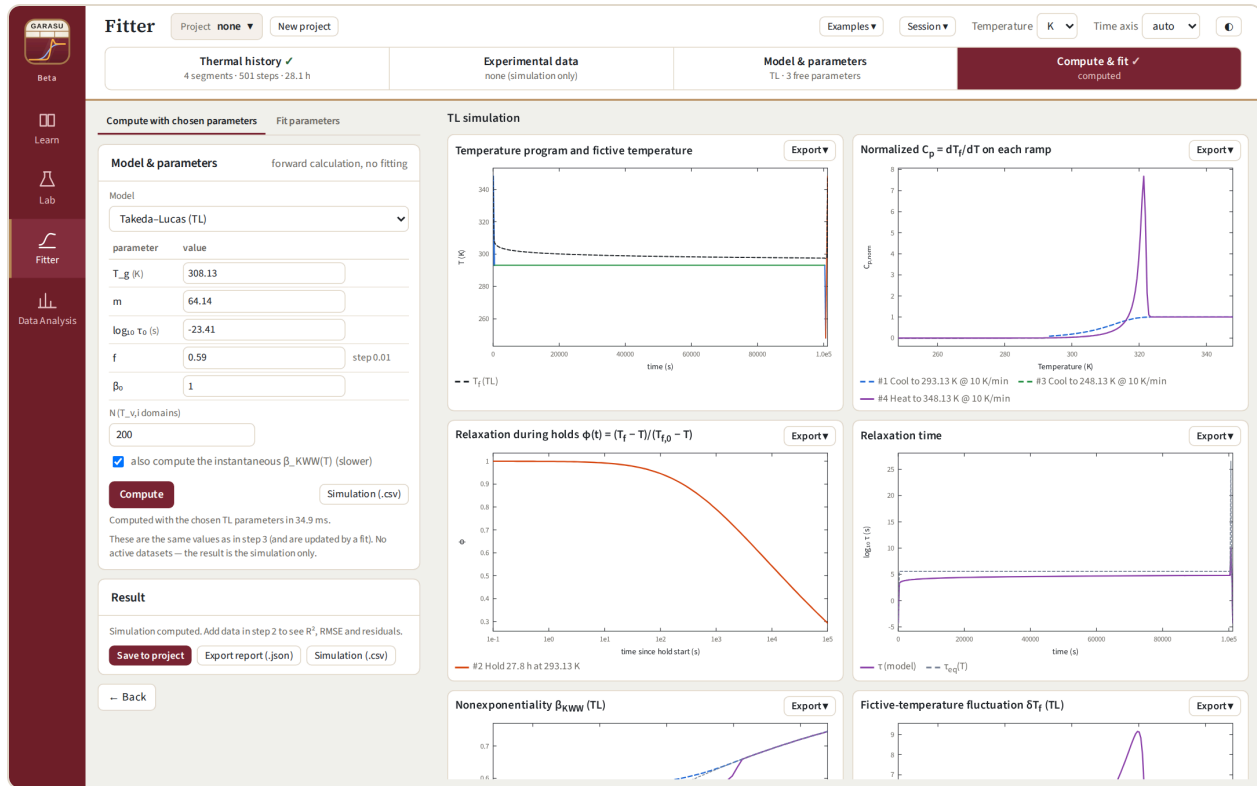


Figure 6.4. Result of **Compute**: temperature program and  $T_f$ , normalized  $C_p$  on each ramp, relaxation during holds, and relaxation time.

### Extra results for the TL model

- Fictive-temperature fluctuation  $\delta T_f$  vs temperature, with the equilibrium value.
- Map of the local fictive temperatures  $T_{f,i}$  vs time, rebinned on a uniform  $T_f$  grid (probability density in  $K^{-1}$ ), with  $T(t)$  and  $\langle T_f \rangle(t)$ .
- $\beta_{KWW}(T)$ : tick also compute the instantaneous  $\beta_{KWW}(T)$  (adds about a second).

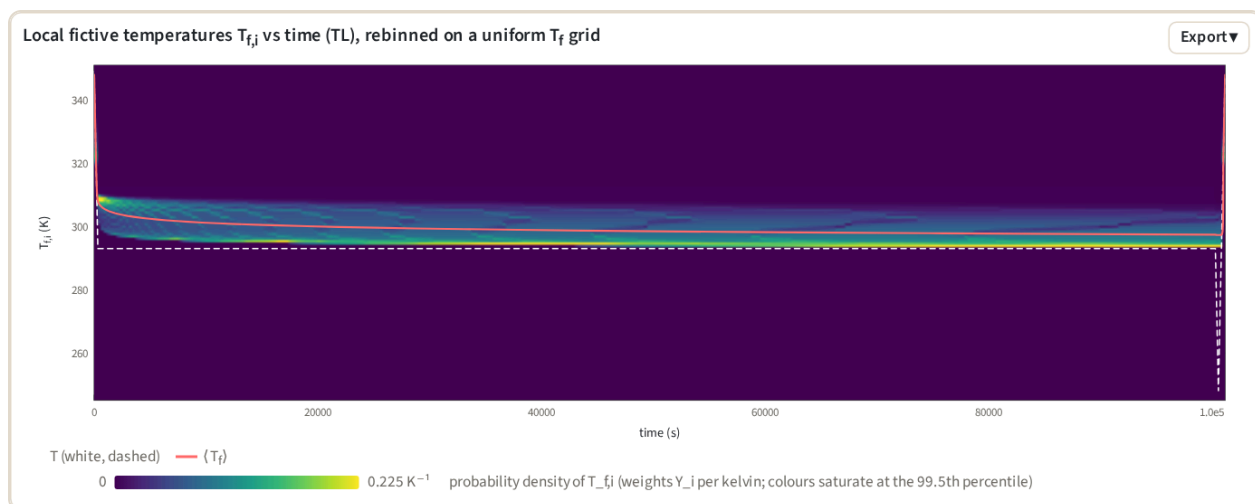


Figure 6.5. TL: local fictive temperatures during cooling, aging and reheating.

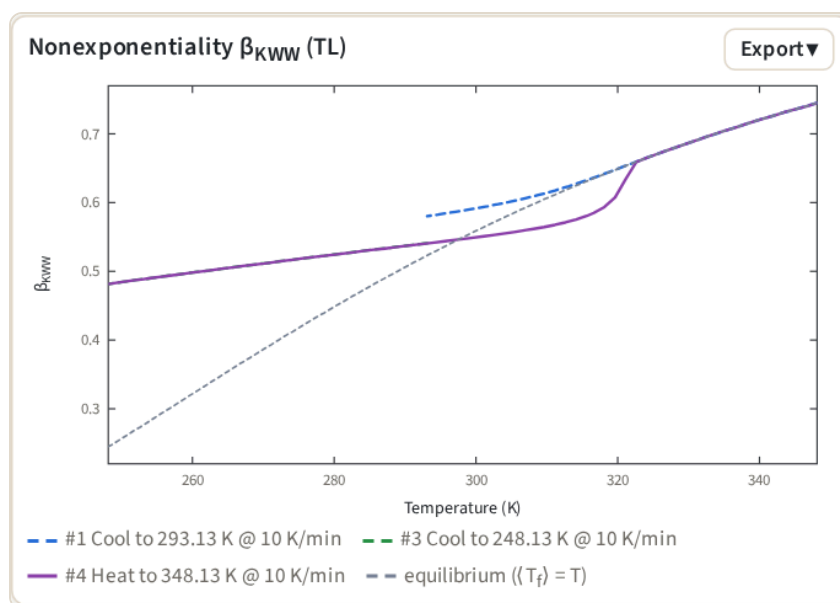
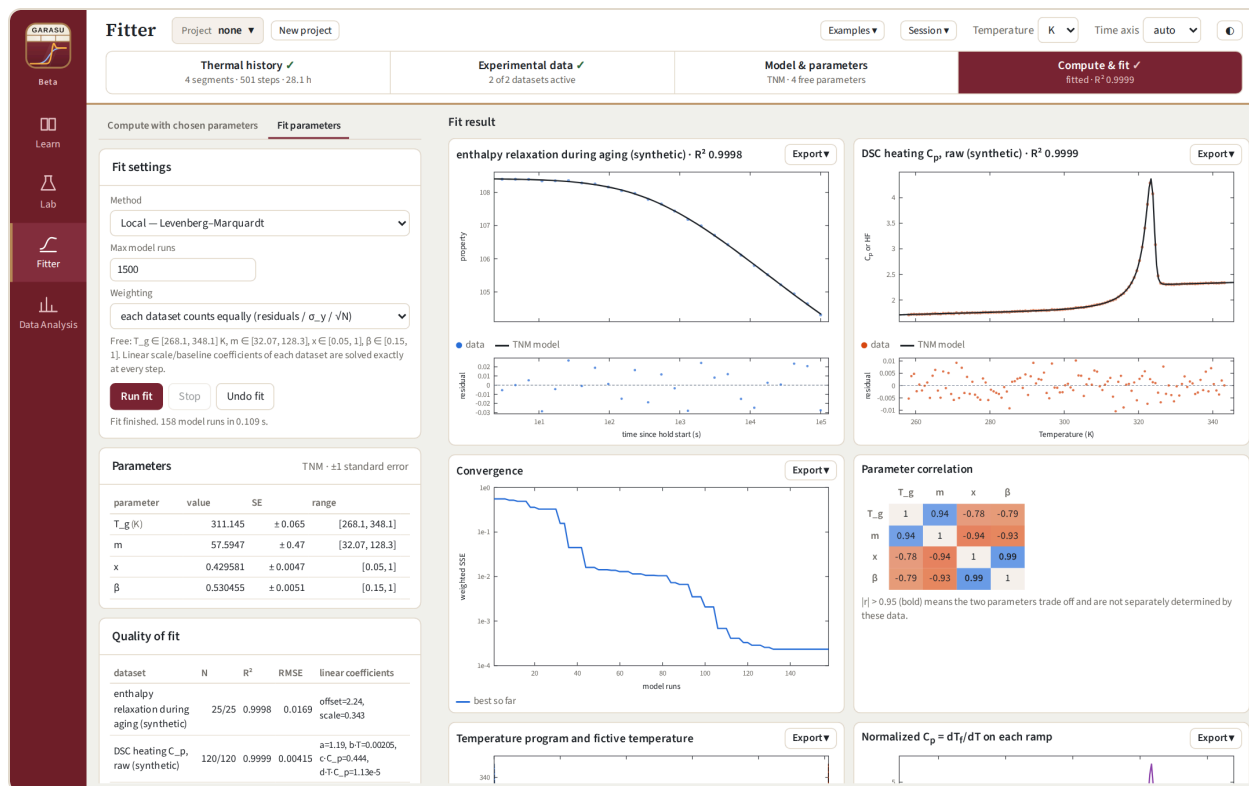


Figure 6.6. TL: instantaneous  $\beta_{KWW}$  along each segment and in equilibrium (grey).

## Fit parameters



**Figure 6.7.** Fit result: data with the fitted model and residuals, convergence, parameter correlation, and the parameter table with standard errors.

- 1 Open the **Fit parameters** tab (needs at least one active dataset and one free parameter).
- 2 Choose the **Method**: *Local - Levenberg-Marquardt* (fast, starts from the current values) or *Global - differential evolution + LM polish* (slower, searches the whole box).
- 3 Set **Max model runs** and the **Weighting** (each dataset equal, or each point equal).
- 4 Click **Run fit**. **Stop** ends early; **Undo fit** restores the previous values.

The result lists each parameter with  $\pm 1$  standard error, flags parameters *at bound* or *poorly determined*, and gives  $R^2$ , reduced  $\chi^2$ , AIC and BIC. In the correlation matrix,  $|r| > 0.95$  (bold) means two parameters trade off and are not separately determined by your data.

### Exports

**Export report (.json)** (parameters, errors, statistics, history), **Model vs data (.csv)**, **Simulation (.csv)** ( $t$ ,  $T$ ,  $T_f$ ,  $\tau$ ,  $\tau_{eq}$ , segment), and **Export ▼** on each plot. **Save to project** stores the whole result in the open project (section 8.3).

## 6.5 Sessions and examples

**Session ▼** holds the Fitter's own state: *New session*, *Save session file (.json)* (history, data, model, values, bounds, fit options) and *Open session file...* Session files from earlier versions ("project" files) open the same way. To keep data and fits of one study together, use a project (chapter 8). **Examples ▼** loads ready-made cases: a selenium DSC cycle, aging + DSC with synthetic enthalpy data, and MDSC with synthetic  $C_p'/C_p''$ .

# 7 • Data Analysis

Data Analysis prepares your own measurements. It has three tabs, shown as a switch in the header. Each tab has **Load example data** (synthetic selenium data made with the TL model, parameters of Se from Takeda & Lucas 2024) so you can try everything first.

## 7.1 Heat-flow scans

Add a heat-flow run

File (.csv, .txt, instrument export)

Choose File

No file chosen

...or paste columns

Time (min),Temperature (°C),Heat Flow (W/g)

0.00,-120.0,-0.1000

0.05,-119.5,-0.1000

0.10,-119.0,-0.1000

0.15,-118.5,-0.1000

...

161 numeric rows, 3 columns detected.

Temperature column

Heat-flow column

Temperature unit

Heat-flow unit

Sign convention

Scan

Rate |q| (K/min)

Annealing time t<sub>a</sub> (s)

Annealing T<sub>a</sub> (K)

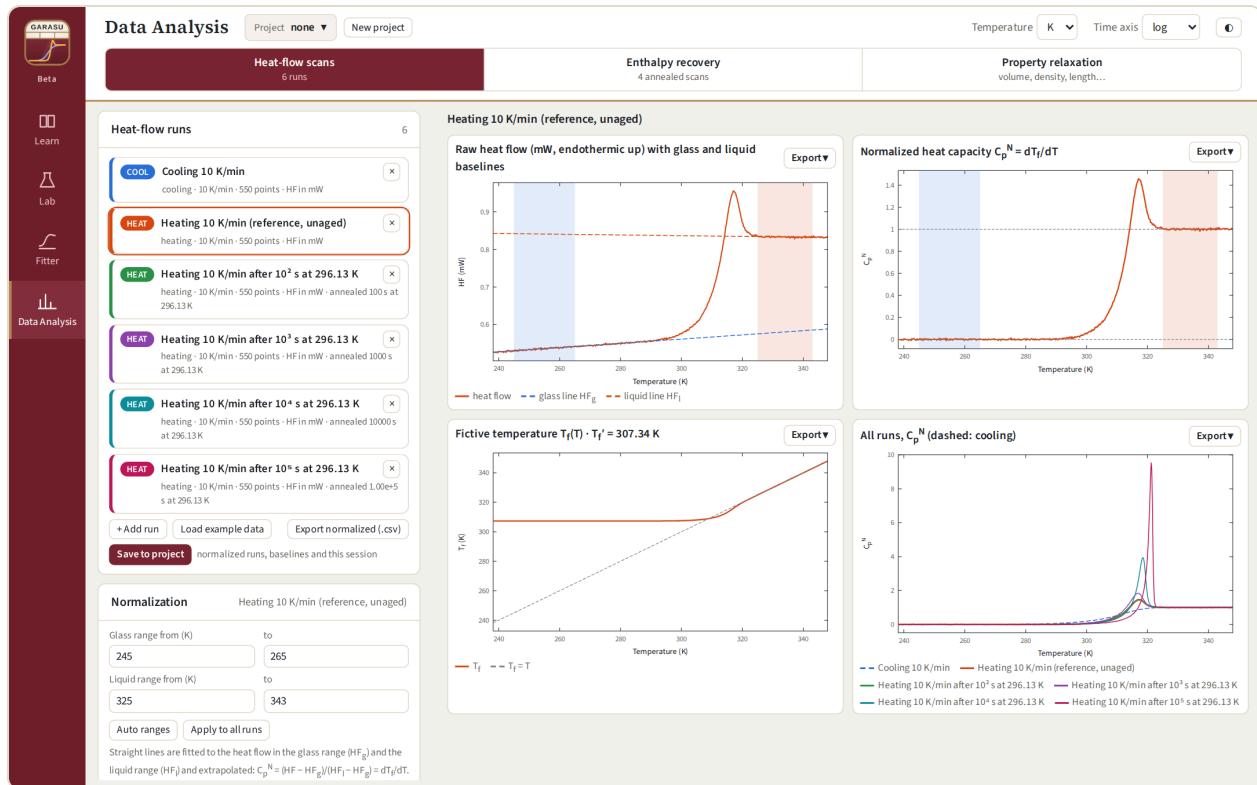
Name

Add run

Cancel

Figure 7.1. Adding a run: paste or load a file, then check the detected columns and units.

- 1 Click **+ Add run**, then choose a file (.csv, .txt, instrument export) or paste columns.
- 2 Check the **Temperature** and **Heat-flow** columns and their units (K or °C; mW, W/g or arbitrary). Column names like “Temperature (°C)” and “Heat Flow (W/g)” set these automatically.
- 3 Set the **Sign convention**: *endothermic up*, or *endothermic down (exo up)* as in TA Instruments exports.
- 4 **Scan** is detected from the temperature trend (heating/cooling). Enter the **rate** and, for mW, the **sample mass**.
- 5 For a heating scan measured after annealing, enter the **annealing time**  $t_a$  and  $T_a$ . Leave them empty for the unaged reference.
- 6 Click **Add run**.



**Figure 7.2.** Normalization of the reference heating scan. Blue band: glass range; orange band: liquid range; dashed: the fitted baselines.

**Normalization.** Enter a **glass range** and a **liquid range** (or **Auto ranges**). A straight line is fitted to the heat flow in each range and extrapolated, giving

$$C_p^N = (HF - HF_g) / (HF_l - HF_g) = dT_f/dT, \quad T_f(T) = T^* - \int_T^{T^*} C_p^N dT'$$

with  $T^*$  at the start of the liquid range (Moynihan's area matching). The card reports the limiting fictive temperature  $T_f'$ , the midpoint ( $C_p^N = 0.5$ ) and  $\Delta C_p$  at  $T_f'$ . **Apply to all runs** copies the ranges; **Export normalized (.csv)** saves  $T$ ,  $HF$ ,  $C_p^N$ ,  $T_f$  for every run; **Save to project** stores them, with both baselines, in the open project (section 8.2).

## Choosing the ranges

- Glass range: well below the transition, where the scan is straight; avoid the start-up transient.
- Liquid range: above the overshoot, where the scan is straight again.
- For comparisons (cooling rates, aging series), use the same ranges for all runs.

## 7.2 Enthalpy recovery

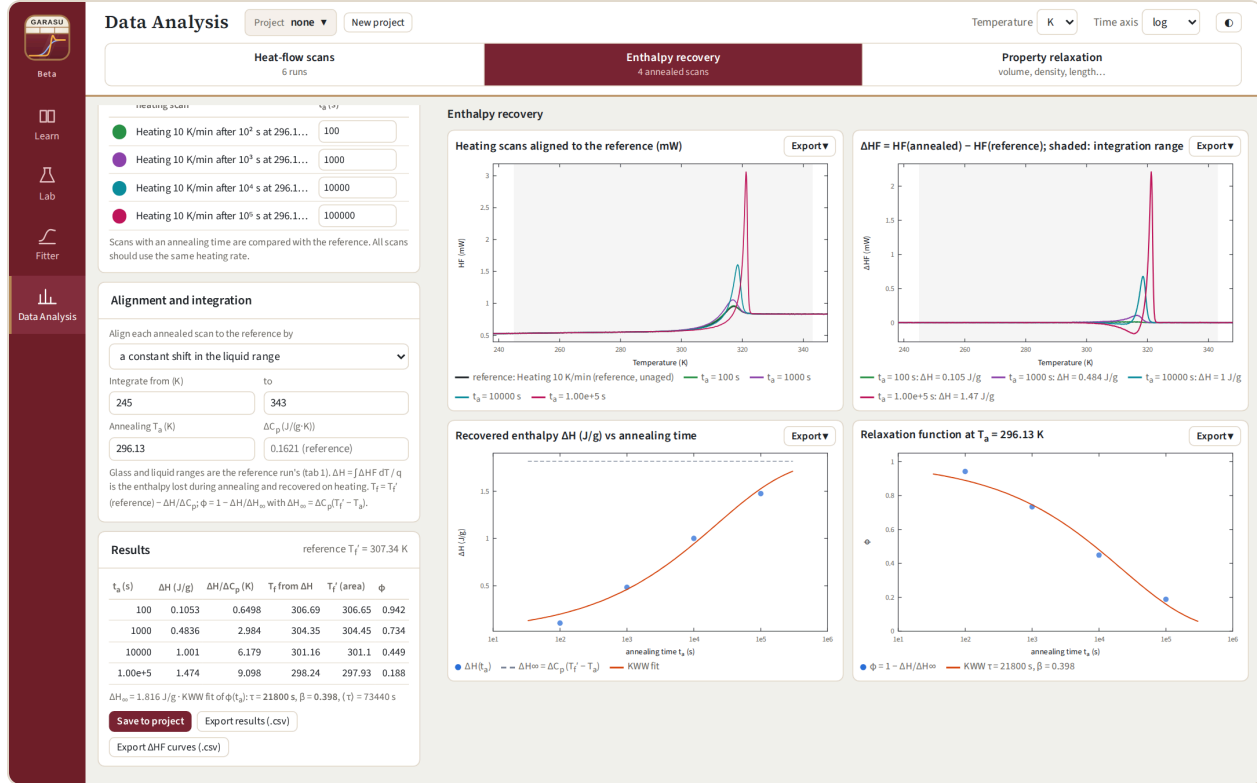


Figure 7.3. Enthalpy recovery: aligned scans, ΔHF(T), recovered ΔH vs annealing time, and φ(t<sub>a</sub>) with a KWW fit.

- 1 Choose the **Unaged reference** (a heating scan without annealing).
- 2 Enter or check the annealing time t<sub>a</sub> of each annealed heating scan.
- 3 Choose how each annealed scan is **aligned** to the reference: a constant shift in the liquid range, or a straight line through the glass and liquid ranges (use this if the glass lines differ slightly).
- 4 Check the **integration range** (default: from the glass range to the liquid range) and enter T<sub>a</sub>.
- 5 Optionally override ΔC<sub>p</sub> (default: the reference at T<sub>f</sub>).

$$\Delta H = (1/q) \int \Delta HF dT, \quad T_f = T_f'(\text{ref}) - \Delta H/\Delta C_p, \quad \phi = 1 - \Delta H/\Delta H_{\infty}, \quad \Delta H_{\infty} = \Delta C_p(T_f' - T_a)$$

| Result column                         | Meaning                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta H$                            | enthalpy lost during annealing and recovered on heating (J/g with mW + mass or W/g; mJ with mW and no mass)                        |
| $\Delta H/\Delta C_p$                 | the same in kelvin: the drop of the fictive temperature during annealing                                                           |
| $T_f$ from $\Delta H$ / $T_f'$ (area) | fictive temperature after annealing, from $\Delta H$ and independently from area matching of the annealed scan (they should agree) |
| $\phi$                                | relaxation function; with three or more annealing times a KWW fit gives $\tau$ , $\beta$ and $\langle \tau \rangle$                |

### Use the same heating rate

$\Delta H$  is computed with the reference's rate. All scans should be heated at the same rate and ideally measured on the same sample.

## 7.3 Property relaxation

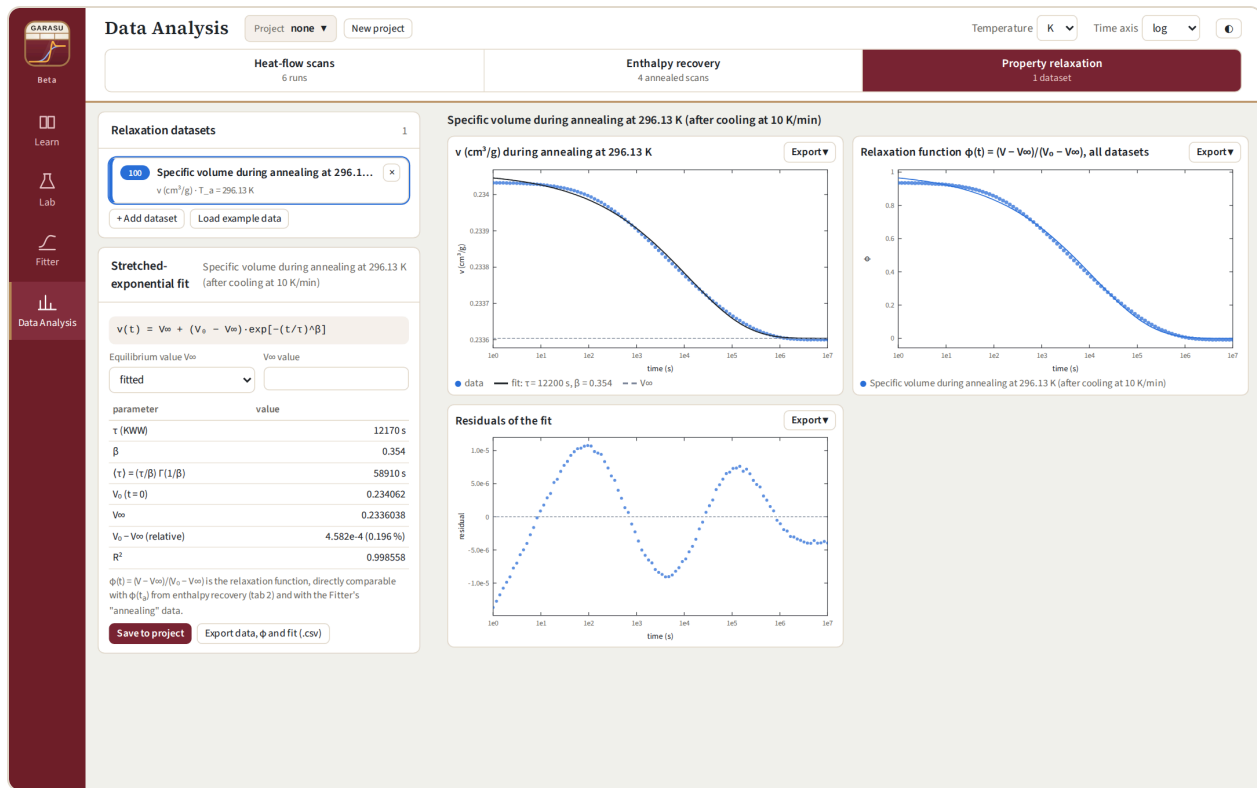


Figure 7.4. Specific volume during annealing with the stretched-exponential fit,  $\phi(t)$  and residuals.

Add a dataset of a property against time during isothermal annealing: volume, density, length, refractive index... (+ Add dataset: pick the time column and unit, the property column, a label and  $T_a$ ). The fit is

$$V(t) = V_{\infty} + (V_0 - V_{\infty}) \exp[-(t/\tau)^{\beta}]$$

with  $V_\infty$  **fitted** or **fixed** at a value you enter (use fixed when the data stop well before equilibrium). The table gives  $\tau$ ,  $\beta$ ,  $\langle \tau \rangle = (\tau/\beta)\Gamma(1/\beta)$ ,  $V_0$ ,  $V_\infty$ , the relative change and  $R^2$ . Several datasets can be compared on the  $\phi(t)$  plot.

Systematic residuals (as in Figure 7.4) are expected for real glasses: relaxation is nonlinear, so a single stretched exponential is only an approximation. Fitting the model in the Fitter accounts for this.

## 7.4 From analysis to fit

The simplest route is a project: save the normalized runs (tab 1) and choose **Fit in the Fitter** in Contents (section 8.3). Alternatively, export the normalized curves (tab 1) or  $\phi(t)$  (tabs 2, 3) as CSV and paste them into Fitter step 2 as “DSC — normalized  $C_p$ ” or “Annealing — normalized relaxation  $\phi(t)$ ”, linked to the matching segment of a thermal history that reproduces your experiment.

## 8 • Projects

A project keeps one study together: the normalized scans from Data Analysis, the fits made from them, and the sessions of both tools. A typical project follows a series of DSC measurements from raw heat flow to fitted model parameters.

| Saved item               | Contents                                                                                                                                                                                                                                                             | Saved from           |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Normalized heat capacity | $T$ , raw HF, $C_p^N$ , $T_f(T)$ ; $T_f'$ , midpoint, $\Delta C_p$ ; scan direction and rate, annealing $t_a$ and $T_a$ ; the glass line $HF_g = a + b \cdot T$ and liquid line $HF_l$ with their ranges                                                             | Data Analysis, tab 1 |
| Enthalpy recovery        | $\Delta H$ , $T_f$ , $\phi(t_a)$ for each annealed scan, and the KWW fit                                                                                                                                                                                             | Data Analysis, tab 2 |
| Property relaxation      | each dataset with its stretched-exponential fit                                                                                                                                                                                                                      | Data Analysis, tab 3 |
| Fit result               | model, all parameters (value, free or fixed, $\pm 1$ standard error, bounds), correlation matrix, $R^2$ , $\chi^2$ , AIC, BIC; thermal history; every dataset with the model curve on the same points, its linear coefficients, and the baselines of its source scan | Fitter, step 4       |
| Session                  | the complete state of the tool, to continue later                                                                                                                                                                                                                    | both tools           |

### Where projects are kept

Projects are saved on this computer, in the browser (web) or in the app (desktop), and appear in every tool and on the home page. The web and desktop versions, and different browsers, keep separate projects: use **Export project file** to move one between them or to back it up (section 8.5).

## 8.1 Create a project

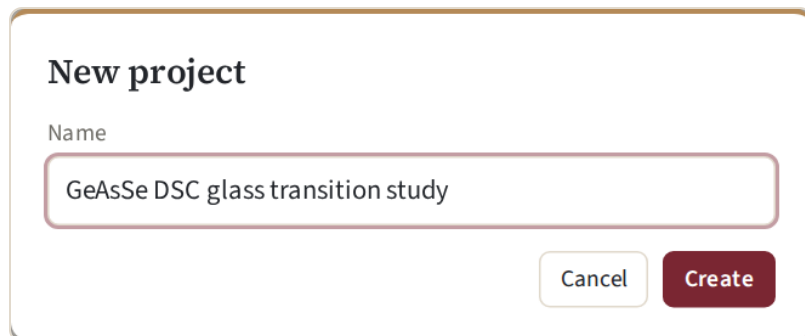


Figure 8.1. New project.

Click **New project** in the header of Data Analysis, the Fitter or the home page, and give it a name, for example *GeAsSe DSC glass transition study*. The project stays open in every tool until you close it or open another. The **Project ▼** menu opens, renames, exports, imports, closes and deletes projects.



Figure 8.2. The project menu.

## 8.2 Save normalized data

- 1 In Data Analysis, add your heat-flow runs and set the glass and liquid ranges (section 7.1).
- 2 Click **Save to project**. Each run is saved as a normalized heat capacity with its baselines, together with the session. Saving again replaces the earlier copy of the same run.
- 3 Tabs 2 and 3 have their own **Save to project** buttons for enthalpy recovery and property relaxation.

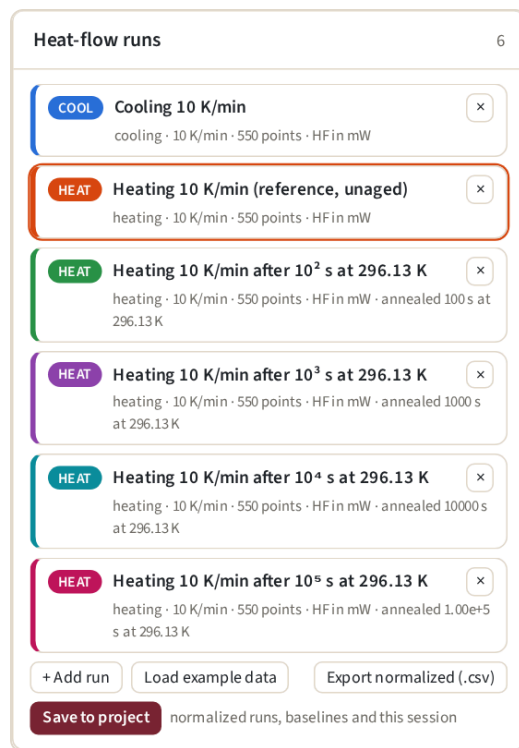


Figure 8.3. Data Analysis, tab 1: **Save to project** below the runs.

## 8.3 Fit from the project

- 1 Open **Contents** and click **Fit in the Fitter** on a normalized run (from Data Analysis) or **Add as dataset** (in the Fitter). Or, in step 2, click + **Add dataset** and choose the **From project** tab.
- 2 Keep **build the thermal history of this scan** checked: the Fitter writes the history from the saved scan (cooling, the annealing hold at  $T_a$  for  $t_a$ , cooling below the data, and the heating scan at its rate) and links the dataset to the scan. Further runs are linked to their own scans.
- 3 Choose the model and free parameters (step 3), then **Run fit** (step 4).
- 4 Click **Save to project** in the results and give the fit a name.

**Add dataset** [X]

Data type  
DSC — normalized  $C_p$  ( $dT_f/dT$ ) vs  $T$  [v]

Linked history segment  
#2 Heat to 348.13 K @ 10 K/min [v]

Paste / file   Synthesize from model   **From project**

Normalized run from "GeAsSe DSC glass transition study"  
Heating 10 K/min after  $10^4$  s at 296.13 K [v]

heating · 10 K/min ·  $T_f = 301.1$  K · annealed 10000 s  
Glass line  $HF_g = 0.3852 + 5.920e-4 \cdot T$  over 245–265 K; liquid line  $HF_l = 0.88539 - 1.556e-4 \cdot T$  over 325–343 K. These travel with the dataset into the saved fit.  
build the thermal history of this scan: cool to 296.13 K, hold  
☒ 10000 s, cool to 238.2 K, heat to 348 K at 10 K/min (replaces the current history)

use  $x \geq$  (K)   use  $x \leq$    weight  
all   all   1

name  
Heating 10 K/min after  $10^4$  s at 296.13 K

☐ fit an offset and a scale factor for this dataset ( $y = a + b \cdot \text{model}$ )

x limits are in K for temperature data and seconds (from the start of the hold) for annealing data.

**Add dataset**

**Figure 8.4.** Fitter, step 2: a normalized run from the project, with its baselines.

The fit result keeps the fitted curve against the experiment point by point, all model parameters with their errors, the quality of fit, and the glass and liquid lines  $HF_g$  and  $HF_l$  of each scan used. Download it from Contents as CSV (data and model columns, with the parameters and baselines in the header) or JSON (everything).

Parameters searched on a grid (such as  $f$  in TL) are free but have no standard error; the CSV marks them as grid-searched.

## 8.4 Contents

**Contents** lists everything saved in the open project, newest first. Each item has an action (**Open here**, **Add as dataset**, **Fit in the Fitter**), **CSV** and **JSON** downloads, and **×** to delete it. **Open here** on a fit result restores its history, data, model and parameters in the Fitter.

## 8.5 Back up and share

**Project ▼** → **Export project file** saves the whole project as one `.garasu.json` file. **Import project file...** adds it on another computer, in another browser, or in the desktop app. Export regularly: clearing the browser's site data removes projects saved in that browser.

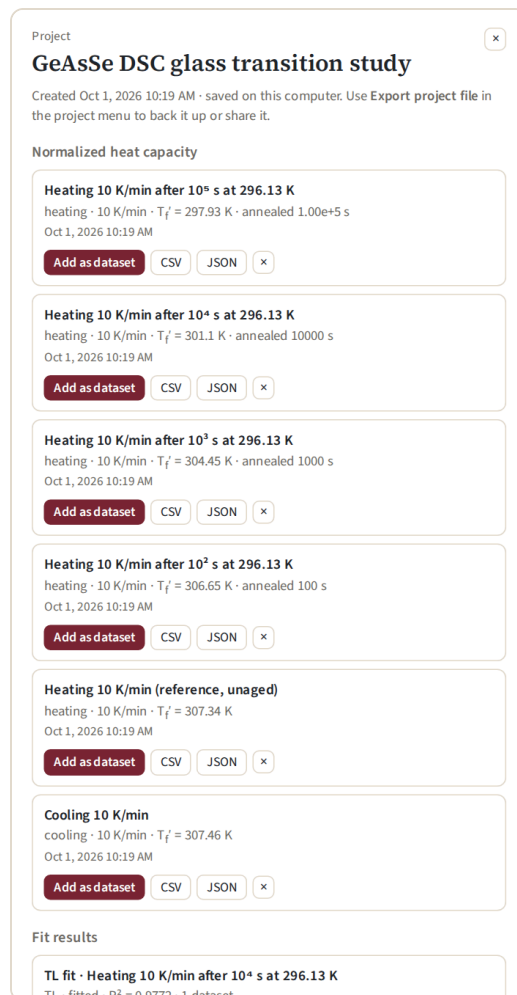


Figure 8.5. Contents of a project, grouped by type.

## 8.6 Export a plot

Every plot in Lab, the Fitter and Data Analysis has **Export ▼**:

| Format           | What you get                                                                                |
|------------------|---------------------------------------------------------------------------------------------|
| PNG / JPEG image | the plot at three times screen resolution, with its title and legend, on a white background |
| SVG (vector)     | the same figure as a vector graphic, to edit in Illustrator or Inkscape                     |
| CSV data         | every curve of the plot as x, y column pairs, with the title and axis labels in the header  |

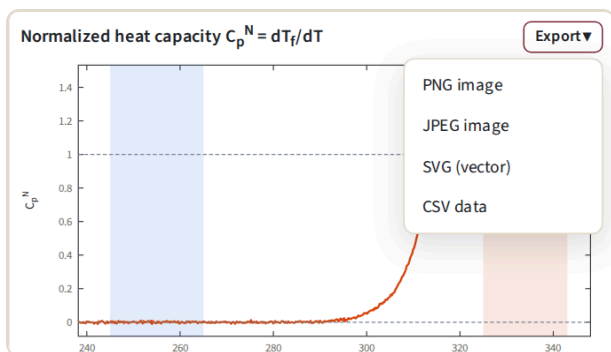


Figure 8.6. Export ▼ on a plot.

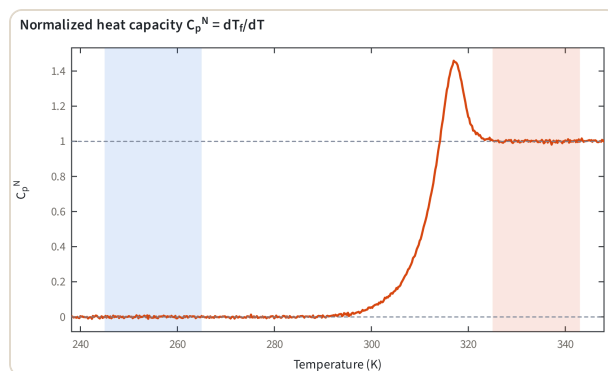


Figure 8.7. The exported PNG.

## 9 • Data files

- **Formats:** text files (.csv, .txt, .dat, .tsv) or pasted text. Columns may be separated by commas, semicolons, tabs or spaces.
- **Headers:** lines before the numbers (titles, column names, unit rows) are skipped; the last text line before the data is used for column names.
- **Decimal point:** use a period ( 0.25 ). Files with decimal commas must be converted first.
- **Order:** any; cooling scans may run from high to low temperature.
- **Instrument exports:** export to text/CSV from the instrument software (e.g. TA Universal Analysis / TRIOS, Netzsch Proteus, Mettler STARE) with temperature and heat flow columns.

### Example (DSC, TA style)

Time (min),Temperature (°C),Heat Flow (W/g)

0.00,-120.0,-0.1000

0.05,-119.5,-0.1001

...

### Example (annealing)

t (s) V (cm<sup>3</sup>/g)

1 0.795716

1.2 0.795715

...

## 10 • Troubleshooting and questions

| Problem                                                          | What to do                                                                                                                                   |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| The Fitter shows no model curves.                                | Expected: the model runs only when you click <b>Compute</b> or <b>Run fit</b> in step 4.                                                     |
| A plot shows “Could not draw ...”.                               | The rest of the page still works. Check the inputs of that part (e.g. a range with fewer than two points) and report it if it persists.      |
| A hold shows no relaxation.                                      | The hold is above $T_g$ : the liquid equilibrates within milliseconds. Put the hold below $T_g$ .                                            |
| Mac: “cannot be opened / cannot verify the developer”.           | System Settings → Privacy & Security → <b>Open Anyway</b> (once per computer).                                                               |
| Mac: “Restart now” does nothing, or the version does not change. | Download the latest DMG and replace the app (section 2.2). Versions before 1.1.6 cannot update themselves.                                   |
| Windows: “Windows protected your PC”.                            | <b>More info</b> → <b>Run anyway</b> (first install only).                                                                                   |
| TL map or $\beta_{KWW}$ plot missing.                            | Very long histories (above ~60 000 steps with 200 domains) do not keep every $T_{f,i}$ . Use a coarser $\Delta T$ step or fewer hold points. |
| A fitted parameter is “at bound”.                                | Widen its bounds in step 3, or fix it if the data cannot determine it.                                                                       |
| Parameters strongly correlated ( $ r  > 0.95$ ).                 | Add data that separate them (several cooling rates, annealing data) or fix one of them.                                                      |
| $C_p^N$ is upside down.                                          | Change the <b>sign convention</b> of the run (endothermic up/down).                                                                          |
| A project is missing.                                            | Projects are kept per browser and per app (section 8.5). Open the browser or app where you made it, or import its exported project file.     |
| Save to project asks for a project name.                         | No project was open. Enter a name to create one and save into it, or cancel and open an existing project from <b>Project ▼</b> .             |
| $\Delta H$ has the wrong sign or size.                           | Check the sign convention, the rate and the sample mass, and that all scans used the same heating rate.                                      |

Report problems or ideas at [github.com/wtakeda-research/GARASU/issues](https://github.com/wtakeda-research/GARASU/issues) or to the Lucas group.

## 11 • Reference

### Model parameters

| Model         | Parameter               | Meaning                                                                                 |
|---------------|-------------------------|-----------------------------------------------------------------------------------------|
| TL            | $T_g$                   | temperature where $\langle \tau \rangle = 100$ s                                        |
|               | m                       | fragility index                                                                         |
|               | $\log_{10} \tau_0$      | high-temperature limit of $\tau$ (s)                                                    |
|               | f                       | width of the relaxation-time distribution ( $\approx \beta_{KWW}$ at $T_g$ ), 0.10–0.99 |
|               | $\beta_0$               | intrinsic (local) stretching; 1 = fully heterogeneous                                   |
|               | N                       | number of domains (numerical; 200 is accurate)                                          |
| TNM           | $T_g$                   | $\tau(T_g) = 100$ s                                                                     |
|               | m                       | fragility; $\Delta h/R = m \ln 10 T_g$                                                  |
|               | x                       | nonlinearity (0–1)                                                                      |
|               | $\beta$                 | KWW stretching exponent                                                                 |
| MAP / RelaxPy | $T_g$                   | $\eta(T_g) = 10^{12}$ Pa·s                                                              |
|               | m                       | fragility (MYEGA)                                                                       |
|               | $\log_{10} \eta_\infty$ | high-temperature viscosity limit                                                        |
|               | A, B, C                 | nonequilibrium viscosity branch; A from continuity at $T_g$ by default                  |
|               | p                       | ergodicity exponent (default $0.3082 \cdot m$ )                                         |
|               | $\log_{10} K_s$         | modulus in $\tau = \eta/K_s$                                                            |
|               | $\beta$                 | stretching of the Prony series                                                          |

## Fitter data types

| Data type                                             | Linked to  | Notes                                        |
|-------------------------------------------------------|------------|----------------------------------------------|
| DSC — normalized $C_p$ ( $dT_f/dT$ ) vs T             | ramp       | already normalized (e.g. from Data Analysis) |
| DSC — $C_p$ or heat flow vs T                         | ramp       | raw; glass/liquid baselines fitted           |
| Enthalpy vs T                                         | ramp, MDSC | $H = a + bT + \Delta C_p T_f$ fitted         |
| Volume / density / length vs T                        | ramp, MDSC | dilatometry                                  |
| Fictive temperature $T_f$ vs T                        | ramp, MDSC |                                              |
| Annealing — normalized $\phi(t)$                      | hold       | $\phi = (T_f - T)/(T_{f,0} - T)$             |
| Annealing — $T_f(t)$                                  | hold       |                                              |
| Annealing — enthalpy / volume / refractive index vs t | hold       | $a + b \cdot T_f$ fitted                     |
| MDSC — $C_p'$ (reversing), $C_p''$                    | MDSC       | normalized; optional scale                   |

## Glossary

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| $T_f$        | fictive temperature: the temperature at which the liquid would have the current structure |
| $T_f'$       | limiting fictive temperature of the glass (frozen value after cooling)                    |
| $C_p^N$      | normalized heat capacity, 0 in the glass and 1 in the liquid; equals $dT_f/dT$            |
| $\phi(t)$    | normalized relaxation function, $1 \rightarrow 0$ during annealing                        |
| $KWW$        | stretched exponential $\exp[-(t/\tau)^\beta]$                                             |
| $\delta T_f$ | spread (standard deviation) of the local fictive temperatures in the TL model             |

## 12 • How to cite, and references

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If GARASU contributes to your work, please cite:

- W. Takeda and P. Lucas, “A model of heterogeneous undercooled liquid and glass accounting for temperature-dependent nonexponentiality and enthalpy fluctuation,” *J. Chem. Phys.* **160**, 174504 (2024). doi:10.1063/5.0196812
- W. Takeda and P. Lucas, “Relationship between enthalpy fluctuation and nonexponential relaxation in glass-forming liquids,” *Mater. Adv.* **7**, 5729 (2026). doi:10.1039/d6ma00442c

and, when you use them, the TNM and MAP / RelaxPy papers:

- A. Q. Tool, *J. Am. Ceram. Soc.* **29**, 240 (1946); O. S. Narayanaswamy, *J. Am. Ceram. Soc.* **54**, 491 (1971); C. T. Moynihan et al., *J. Am. Ceram. Soc.* **59**, 12 (1976); I. M. Hodge, *Macromolecules* **20**, 2897 (1987); G. W. Scherer, *J. Am. Ceram. Soc.* **69**, 374 (1986).
- J. C. Mauro, D. C. Allan and M. Potuzak, *Phys. Rev. B* **80**, 094204 (2009); X. Guo et al., *J. Am. Ceram. Soc.* **101**, 1169 (2018); J. C. Mauro and Y. Z. Mauro, *Physica A* **506**, 75 (2018); J. C. Mauro, R. J. Loucks and P. K. Gupta, *J. Am. Ceram. Soc.* **92**, 75 (2009); C. J. Wilkinson, Y. Z. Mauro and J. C. Mauro, *SoftwareX* **7**, 255 (2018).

Experiment references (Kovacs, Macedo–Napolitano, Hodge, Schawe, Hutchinson–Montserrat) are listed in Learn → Further reading.

GARASU • Lucas group, Department of Materials Science and Engineering, University of Arizona • [github.com/wtakeda-research/GARASU](https://github.com/wtakeda-research/GARASU)